

TECHNICAL RESPONSE

Response to the Independent Technical Review and Feasibility Assessment

Scientific judgment, adoption decisions, and recommended Version 3 direction for iPIN-OpenPPI

Executive verdict

The independent review is strong, technically credible, and constructive. It does not weaken the iPIN-OpenPPI concept; it identifies where the original blueprint was insufficiently operational, statistically under-specified, or too broad for one critical path. Its principal recommendations should be adopted.

Prepared for

The iPIN-OpenPPI expert working group

Basis of assessment

Independent Technical Review and Feasibility Assessment, 2 August 2026

Status: working-group discussion document

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Source note. *This response is grounded in the supplied “iPIN-OpenPPI Independent Technical Review and Feasibility Assessment” dated 2 August 2026. Literature references mentioned here are those cited in that review; this document is an assessment and action response, not a separate full literature re-audit.*

1. Overall judgment

This is a strong, technically credible, and constructive independent review. It does not invalidate the iPIN-OpenPPI concept. Rather, it identifies the points at which the original blueprint was scientifically overambitious, insufficiently operational, or statistically under-specified, and it converts the programme into a more defensible research plan.

I agree with most of its major conclusions. In particular, the review correctly recognizes that the project's strongest contribution is not merely a new neural-network architecture. The durable contribution is the integrated treatment of evidence provenance, assay-specific measurement, uncertain negative labels, non-random selection of tested pairs, leakage-resistant evaluation, calibration, selective prediction, and prospective validation.

The review's proposed repositioning is therefore excellent and should become the central programme description.

Recommended programme positioning

An evidence- and evaluation-first programme for direct protein-protein interaction prediction under assay and selection bias, using partner-aware sparse sequence modeling and prospective validation.

This formulation is scientifically more precise and likely to be more compelling to expert reviewers than a claim that the project estimates one universal, sequence-derived probability that any two proteins interact.

Why the review is valuable

- It protects the programme from making a probability claim that cannot be falsified or calibrated.
- It corrects an incomplete assay-observation model that allowed false negatives but not false positives.
- It identifies a genuine architectural weakness: partner-independent interface routing.
- It reduces the critical path before architectural and wet-lab risks accumulate.
- It preserves the highest-value contributions: evidence semantics, strict evaluation, calibration, and prospective validation.

2. What the review changes

The review should be treated as a successful scientific red-team exercise. Its function is not to replace the blueprint, but to distinguish three categories of content: principles that should be retained, assumptions that must be corrected, and attractive extensions that should be removed from the initial critical path.

Category	Assessment	Action
Retain	Evidence records, separation of direct binding from association, strict leakage controls, realistic prevalence, calibration, phased decision gates.	Keep as core programme principles.
Correct	Universal intrinsic probability, one-sided assay model, omitted pair-selection process, partner-independent routing, C2 definition.	Revise before implementation begins.
Defer	Mechanism mixture of experts, three-protein competition, broad affinity regression, large strict-PLM pretraining, simultaneous teacher branches.	Move outside the first 24-month critical path.

The best response is therefore not to defend every element of the original document. It is to incorporate the review's strongest corrections into Version 3 while preserving the original programme's evidence-first identity.

3. Primary target and estimand

3.1 The criticism of “intrinsic interaction probability” is correct

The original target - the intrinsic ability of two canonical sequences to form a direct interaction under some compatible biochemical condition - is biologically intuitive but not yet operationally measurable.

The phrase “under some compatible condition” makes the target existential. A nominally negative pair may bind only in a particular post-translational state, isoform, oligomeric state, concentration range, membrane environment, ligand-induced conformation, cleavage state, or construct configuration. Consequently, a universal negative label is almost impossible to establish.

The review is therefore right that the output cannot initially be presented as a universally calibrated biochemical probability.

A more defensible target is a probability averaged over an explicit reference distribution of biochemical contexts:

$$p_{ref}(A,B) = E_{\{c \sim q_{ref}\}} [P(\text{direct binding} \mid S_A, S_B, c)]$$

Conceptually, this is sound. In practice, defining a defensible q_{ref} across heterogeneous proteins may itself be difficult. The project should therefore avoid making the success of its first model depend on a fully universal reference distribution.

3.2 Recommended initial outputs

The first model should expose two clearly distinguished outputs:

1. An assay- and construct-conditioned interaction probability: $P(y = 1 \mid S_A, S_B, \text{assay, constructs, orientation, recorded context})$.
2. An evidence-integrated sequence compatibility score intended for ranking, transfer, and prioritization.

The second output may become a calibrated latent probability only if synthetic recovery, prior-sensitivity, and cross-assay identifiability experiments support that interpretation. Until then, “compatibility score” is the scientifically safer term.

Recommended language

Do not claim that the first model estimates a context-free biochemical truth. Claim that it predicts direct-interaction evidence under a defined assay/search space and produces a sequence-derived compatibility score whose latent interpretation is explicitly tested.

4. Observation model, assay errors, and selection bias

4.1 Adding assay false positives is essential

The original observation model effectively assumed that assays could miss a true interaction but could not produce an artifactual positive. That assumption is not defensible for heterogeneous PPI evidence.

Observed positives can arise from autoactivation, indirect bridging, contamination, construct-induced binding, mapping errors, incorrect biological-assembly assignments, or curation mistakes. The observation model therefore requires both sensitivity and specificity.

$$P(y = 1 | b, m, x) = b \text{ Se}_m(x) + (1 - b) [1 - \text{Sp}_m(x)]$$

Here, b denotes binding in a specified context, m denotes the assay, and x contains orientation, construct design, expression success, abundance, or other recorded features. Orthogonally supported positives and strong non-binding evidence should act as high-reliability probabilistic anchors rather than deterministic truth.

4.2 Identifiability is the central statistical risk

Simultaneously learning latent binding, assay sensitivity, assay specificity, context effects, and selection bias from heterogeneous databases may be severely underidentified. The full latent model should therefore be a research hypothesis, not the only implementation path.

The programme should compare nested models on identical data:

- Assay-specific discriminative classifiers.
- Random-negative and matched-negative binary cross-entropy baselines.
- Non-negative positive-unlabeled learning.
- Hierarchical assay heads with shared sequence representations.
- A restricted sensitivity/specificity model fitted only where repeated or cross-assay evidence is sufficient.
- The full latent observation model, evaluated for recovery and prior stability.

If intrinsic and assay components cannot be separated reliably, the correct fallback is not failure. It is calibrated assay-specific prediction plus an evidence-integrated compatibility score.

4.3 The omitted pair-selection process is real

Public PPI databases do not contain a random sample of protein pairs. Testing and publication depend on protein popularity, disease relevance, pathway hypotheses, known domains, clone availability, abundance, localization, assayability, prior predictions, and institutional priorities.

The statistical system should distinguish three processes: whether a pair is selected and successfully tested; whether binding exists under the relevant biochemical context; and whether the assay reports a positive result.

$$t_{ABm} \sim P(t = 1 | A, B, \text{assay}, \text{study history}, \text{assayability})$$

I agree with this diagnosis, but I would not initially attempt to estimate one universal selection model across the entire historical literature. Many study-selection mechanisms are unrecorded and may be unrecoverable. The more practical strategy is to develop the statistical model first on one or more systematic screens with an explicitly defined tested universe, then use heterogeneous curated evidence as supporting or transfer data.

Practical statistical strategy

Use systematic assay matrices as the primary identifiable pilot. Treat the full literature as a biased evidence ecosystem and perform sensitivity analyses rather than claiming that all historical selection mechanisms have been reconstructed.

5. Architecture: partner-aware sparse routing

5.1 The partner-independent router should be replaced

This is the strongest architectural criticism in the review. A protein may contain several potential interfaces and use different domains, motifs, or disordered regions for different partners. A router that sees only protein A can learn generic interface propensity, domain boundaries, conserved patches, or frequently used surfaces, but it cannot reliably decide which region of A is relevant specifically to protein B.

The routing stage should therefore select joint patch pairs rather than independent top-k regions.

Recommended routing design:

1. Encode both chains with shared sequence-encoder weights.
2. Pool residue embeddings into overlapping coarse patches at motif and domain scales.
3. Compute an inexpensive partner-conditioned patch-pair compatibility matrix.
4. Select joint patch pairs rather than independent per-protein regions.
5. Apply high-resolution sparse cross-attention only to selected patch pairs.
6. Retain a global interaction path and a full-attention path for short protein pairs.
7. Aggregate the biological interaction head with exact swap symmetry.

This preserves the computational motivation for sparse local reasoning while making the routing biologically partner-specific.

5.2 Oracle experiments should decide whether routing is worth the complexity

The review's proposed controls are excellent and should be treated as decision experiments rather than routine ablations.

- Full attention versus sparse attention at matched parameter count and compute.
- Fixed windows versus learned routers.
- Independent routers versus joint partner-aware routing.
- Predicted windows versus oracle structural-interface windows.
- Global-only versus local-only versus combined paths.
- Single-scale versus motif-scale plus domain-scale windows.
- Routing stability across seeds and closely related alternative partners.

Observed result	Interpretation
Oracle windows do not improve strict prediction.	Interface-local reasoning is probably not the principal bottleneck; do not invest heavily in routing.
Oracle windows help, learned routing does not.	The proposer or routing objective is the bottleneck, not the cross-attention mechanism.
Oracle and learned partner-aware routing both help.	The sparse-interface hypothesis is supported and merits continued development.

5.3 Multi-task scope should be staged

The original set of heads - interaction, assay observation, type, interface residues, contacts, affinity, mutation effects, mechanism experts, and uncertainty - is too broad for simultaneous activation. These tasks differ in dataset composition, units, conditions, missingness, and leakage risk.

Recommended order:

1. Reference interaction or compatibility score plus assay-observation head.
2. Interface-region supervision.

3. Mutation ranking within coherent scans or complexes.
4. Optional contact distillation.
5. Affinity only on conditionally coherent subsets.
6. Mechanism experts only after label sufficiency and benefit have been demonstrated.

The three-protein competition/cooperation extension should remain outside the initial critical path.

6. Scope, staffing, and programme feasibility

The review is correct that the complete original programme is overloaded for a single 24-month critical path. The following branches cannot all be executed rigorously in parallel without a sizeable consortium: exhaustive construct curation, several strict benchmarks, latent selection-aware statistics, sparse architecture, contact and mutation learning, affinity regression, structure and MSA distillation, mechanism experts, strict PLM pretraining, retrieval, graph reconstruction, three-protein modeling, prospective screening, and orthogonal validation.

6.1 Recommended deferrals

- Mechanism mixture of experts.
- Three-protein competition and cooperation.
- Broad cross-study affinity regression.
- Large custom PLM pretraining as a flagship objective.
- Simultaneous structural and MSA distillation before either teacher branch has passed a gate.

6.2 Staffing interpretation

The review's estimate of roughly six to ten computational and biological full-time equivalents is reasonable for the complete original scope, especially when extensive curation, statistical leadership, HPC engineering, and wet-lab validation are included.

A focused programme could be delivered by a smaller team if the critical path is reduced. A realistic minimum structure would include:

- One strong data and benchmark engineer.
- Two model researchers or research engineers.
- A curator or protein scientist with protected time.
- Part-time statistical and HPC/MLOps support.
- A committed external experimental group with agreed assay throughput and orthogonal follow-up.

This remains a substantial project, but it does not necessarily require ten internal full-time staff if the initial scope is disciplined and external experimental capacity is genuinely committed.

7. Benchmark and evaluation recommendations

7.1 Correct the C1/C2/C3 definitions

Category	Exclusive definition
C1	Both proteins have appeared in training, but the specific pair is unseen.
C2	Exactly one member of the test pair is unseen in training.
C3	Both proteins in the test pair are unseen in training.

This correction matters because defining C2 as “at least one protein unseen” makes C3 a subset of C2 and obscures the true difficulty ladder.

7.2 Use a benchmark ladder, not only one extreme intersection

Intersecting C3, 30% sequence clustering, publication-time holdout, interface-family holdout, assay holdout, and species holdout may leave a small and atypical test set. The programme should report separate axes plus one prespecified combined stress test.

Benchmark axis	Primary purpose
C1/C2/C3	Protein novelty and pair generalization.
40%, 30%, and 20% sequence clustering	Homology transfer and sensitivity to family-level exposure.
Publication-time split	Retrospective discovery simulation.
Assay/source holdout	Measurement-platform transfer.
Species holdout	Evolutionary transfer.
Interface-family subset	Generalization to novel binding modes on audited structural examples.
Combined stress test	Prespecified worst-case generalization.

For every split, report the surviving numbers of proteins, positive pairs, conditional negatives, sequence families, assays, publications, and interaction mechanisms. Statistical uncertainty must reflect dependence among proteins, families, complexes, studies, and assay batches rather than treating pairs as independent observations.

7.3 Unknown proteome pairs are not verified negatives

Proteome-scale precision cannot be estimated by declaring all undocumented pairs false. This would count undiscovered positives as false positives and produce an uninterpretable calibration claim.

Reliable precision estimates require one of the following:

- A systematic tested universe with known positive, negative, failed, and untested outcomes.
- A strict later-evidence evaluation.
- Prospective experimental validation.
- Sensitivity analysis across plausible prevalence and hidden-positive rates.

Calibration must therefore always be tied to a defined organism, candidate universe, assay, retrieval policy, and assumed prevalence. It is not one universal property of the model.

7.4 Model selection should use one primary metric plus hard constraints

The review is right to reject an opaque geometric mean of metrics evaluated under different prevalences. The project should prespecify one primary decision metric, impose hard minimum requirements for precision and calibration, and use a secondary dashboard for coverage, efficiency, mechanism balance, and uncertainty behavior.

8. Evidence warehouse and negative-label semantics

8.1 Construct resolution may dominate the curation workload

The review correctly predicts that construct-to-sequence mapping may be the most difficult data-engineering component. Relevant issues include fragments, domains, isoforms, mutations, tags, fusion proteins, signal-peptide or propeptide removal, ambiguous numbering, supplementary-method sequences, orientation, and technical expression status.

Construct-mapping completeness should be reported as a first-class metric. A smaller warehouse with reliable construct provenance may be scientifically more valuable than a much larger collection of nominal canonical protein pairs.

8.2 Negative evidence should remain conditional

Correct interpretation of most negative records

No interaction was detected for these constructs, in this orientation, under these conditions, using this assay. This is not equivalent to proving that the canonical sequences can never bind.

Recommended evidence categories include technical failure, assay-negative with complete technical checks, assay-negative with incomplete metadata, orthogonally supported non-binding under specified conditions, biological-incompatibility hypothesis, and untested/unlabeled. These categories can receive different likelihood weights without being collapsed into hard universal negatives.

8.3 Structural evidence requires independent quality controls

PDB-derived contacts are not automatically biological direct interactions. Crystal packing, engineered assemblies, antibodies, repeated subunits, uncertain biological units, and chain-mapping errors must be handled explicitly. Structural records should preserve biological-assembly provenance, interface size and quality, method and resolution, mapping confidence, symmetry, oligomeric state, and engineered-construct status.

8.4 Licensing should be an explicit deliverable

Before broad ingestion, the team should build a source-by-source licensing and redistribution matrix covering raw records, transformed records, identifiers, sequences, derived labels, commercial-use restrictions, required attribution, model-weight redistribution, and pseudo-label or teacher-output release. This may determine whether the public product is a complete warehouse or reproducible ingestion code plus versioned manifests.

9. Arrhenius implementation assessment

The review's hardware conclusion is sound: Arrhenius has sufficient aggregate GPU memory, CPU capacity, local NVMe, shared storage, and interconnect for the proposed workloads. The uncertainty is not theoretical hardware capacity; it is the project's actual allocation and measured scaling efficiency.

9.1 Early engineering risks

- aarch64 compatibility for PyTorch extensions and specialized scientific software.
- FlashAttention, xFormers, and fused-optimizer support.
- FAISS GPU and multi-vector retrieval implementation.
- MMseqs2 and structural-alignment packages.
- Apptainer behavior for multi-node jobs.
- NCCL and Slingshot scaling.
- Distributed checkpointing and recovery.
- Realistic throughput across short, median, long, and pathological sequence pairs.

9.2 A compute formula is not yet a compute budget

A serious allocation application should include measured rather than nominal quantities: parameter and trainable-parameter counts, pair-token totals, sequence-length distribution, selected-patch distribution, tokens per second per GPU, optimizer steps, evaluation and teacher overhead, seed and ablation counts, checkpoint and embedding storage, and one-node versus multi-node efficiency.

Recommended profiling gate

Profile the same representative workloads on one and four GH200 GPUs before requesting a large allocation. Use those measurements to separate mandatory experiments from optional branches.

10. Literature positioning and novelty

The review is correct that recent work narrows broad architectural novelty claims. Partner-aware sequence modeling, pair-language models, local interaction grammars, uncertainty-aware prediction, and residue-level pair context already exist. MSA Pairformer, SPPIDER-seq, and ReCLIP should be incorporated into the landscape, along with the reference corrections identified in the review.

These works do not invalidate iPIN-OpenPPI. They change where novelty should be claimed.

Strongest novelty claim

The coherent combination of evidence-record semantics, assay and selection modeling, partner-aware sparse local reasoning, strict leakage audits, calibrated candidate selection, and prospective validation. Architecture alone is unlikely to support the strongest claim.

A strict custom PLM should be treated as a small audit model that estimates the contribution of public pretraining exposure. It need not outperform the public backbone. Its scientific role is exposure analysis, not scale competition.

11. Version 3 adoption decisions

Blueprint component	Decision	Implementation meaning
Evidence-record warehouse	Adopt unchanged	Retain as the foundational data object.
Direct binding separated from association	Adopt unchanged	Keep direct binary, co-complex, and functional evidence distinct.
Strict split ladder	Adopt with correction	Use exclusive C2 and report separate axes plus one stress test.
Universal intrinsic probability	Replace	Use a conditional estimand or compatibility score until identified.
Assay observation model	Expand	Include sensitivity, specificity, and technical/design covariates.
Pair-selection model	Pilot selectively	Develop first on systematic screens; avoid global claims.
Independent interface router	Replace	Use partner-aware joint patch-pair routing.
Contact and mutation heads	Stage	Introduce only after the core model passes strict gates.
Affinity regression	Restrict	Use coherent condition-specific subsets and within-study ranking.
Mechanism mixture of experts	Defer	Remove from the initial critical path.
Three-protein extension	Defer	Treat as a later independent project.
Strict PLM track	Narrow	Train a small exposure-audit model.
Prospective experiment	Start early	Secure assay, partner, throughput, and controls at kickoff.

12. Recommended six-month de-risking objective

At month six, the programme should have produced a scientifically useful result even if the universal latent model proves impossible. The de-risking phase should target the following concrete outcomes:

1. An operational definition of the primary target and its reference assay or context.
2. One systematic evidence subset with a known tested universe.
3. Construct-mapping and metadata-completeness statistics.
4. Frozen C1, exclusive C2, C3, sequence-cluster, and temporal splits.
5. Metadata, interolog, bi-encoder, and simple joint-encoder baselines.
6. A synthetic identifiability study with known sensitivity, specificity, and selection mechanisms.
7. A controlled comparison of BCE, matched-negative, PU, and restricted assay-reliability models.
8. Oracle-window, independent-router, and partner-aware routing experiments.
9. Measured one- and four-GH200 throughput on representative pair-length regimes.
10. A committed assay, laboratory partner, candidate budget, and orthogonal-validation route.

12.1 Recommended continuation gates

Gate	Proceed only if...
Evidence	Constructs and technical failures are sufficiently resolved; strict splits retain multiple independent families and mechanisms.
Statistical	The latent model is recoverable or stable under plausible priors and improves strict ranking or calibration over simpler baselines.
Routing	Oracle regions help, learned partner-aware routing captures a meaningful fraction of that benefit, and sparse computation reduces long-pair cost.
Retrieval	Recall at the intended candidate budget is high enough that reranking is not retrieval-limited.
Experimental	Assay throughput, construct route, controls, technical-failure handling, and orthogonal validation are secured.

If the statistical gate fails, the project should narrow the target to assay-specific calibrated prediction and evidence integration rather than forcing an uninterpretable universal latent probability. If the routing gate fails, retain the evidence and benchmark programme and use a simpler joint encoder.

13. Final verdict

I would accept the principal recommendations of the independent review. It identifies three genuine weaknesses in the original blueprint: an insufficiently operational biological target, an incomplete evidence-generation model, and a partner-independent interface router.

It also prevents the programme from becoming an accumulation of interesting but weakly integrated subprojects. The strongest response is a Version 3 centred on a narrower and falsifiable claim: predict and prioritize direct heteromeric protein interactions from sequence while explicitly accounting for assay reliability, conditional negative evidence, selection bias, and strict biological novelty.

With that revision, the project becomes more feasible, more falsifiable, and more publishable. The independent review strengthens the programme substantially rather than diminishing it.

Recommended final decision

Approve the six-month de-risking phase. Do not commit at kickoff to every auxiliary head, teacher, mechanism expert, strict pretraining branch, triplet extension, or large prospective campaign. Continue the full programme only after evidence, statistical, routing, compute, retrieval, and experimental gates have been met.

Proposed concise programme statement

iPIN-OpenPPI will develop evidence-aware, assay- and selection-conscious sequence models for prioritizing direct heteromeric protein interactions, evaluated under strict biological novelty and validated prospectively where feasible.

Reference basis

1. iPIN-OpenPPI Independent Technical Review and Feasibility Assessment. Prepared for the iPIN-OpenPPI expert working group, 2 August 2026.
2. The independent review's selected literature list, including PLM-interact, paired sequence language modeling, MINT, SWING, DataSAIL, the PLM leakage analysis, experimental assessment of AI-based interactome mapping, SPPIDER-seq, MSA Pairformer, ReCLIP, and official NAISS Arrhenius documentation.

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